

CS 521: Introduction to Post Quantum Security

Course Code: CS 521

Course Name: Introduction to Post Quantum Security

L-T-P-C: 2-1-0-3

Prerequisites: DS 404

Students intended for: M.Tech/Ph.D./B.Tech 3rd/4th year

Elective or Core: Elective

Mutual Exclusion: -

Approval: 59th BoA

Course Contents

- **Introduction to Cryptography and Security Goals:** Security Goals in Cryptography, Cryptographic Attacks, Cryptographic Services and Mechanisms, Techniques for Implementing Security Goals. [4 Lectures]
- **Pre-Quantum cryptography and its vulnerability:** Symmetric Cryptography: AES, DES, and Hash Function, Asymmetric Cryptography: RSA, ECC, and Diffie-Hellman, How Quantum Computers Threaten Classical Cryptosystems. [8 Lectures]
- **Foundations of Post-Quantum Cryptography (PQC):** What is post-quantum cryptography, Challenges in transitioning to PQC, Overview and Motivation for PQC, The Need for Quantum-Safe Cryptography, Overview of the impact of quantum computing on cryptography [6 Lectures]
- **Post Quantum Algorithms:** FIPS 203- Module-Lattice-Based Key-Encapsulation Mechanism (KEM) Standard, FIPS 204- Module-Lattice-Based Digital Signature Standard, FIPS 205-Stateless Hash-Based Digital Signature Standard. Study of recent protocols: TLS (Transport Layer Security), IPSec (Internet Protocol Security), SSH (Secure Shell), DNSSEC (Domain Name System Security Extensions), Blockchain & Cryptocurrencies, 5G-Authentication Key Agreement (5G-AKA). [10 Lectures].

Text books:

1. Daniel J. Bernstein, Johannes Buchmann, and Erik Dahmen, **Post Quantum Cryptography**.
2. Alex C Techworth, **Quantum Cryptography: Securing the Future with Quantum Tech**.
3. Christof Paar, Jan Pelzl, Tim Güneysu, **Understanding Cryptography: From Established Symmetric and Asymmetric Ciphers to Post-Quantum Algorithms**.

References:

1. <https://researchportal.vub.be/en/persons/an-braeken>
2. <https://www.surrey.ac.uk/people/mohammad-shojafar>
3. https://www.iitr.ac.in/~CSE/Gangopadhyay_Sugata